Assessments

STAT 109 — Summer Session 1, 2026 (online, async)

Assessments

Graded work happens in **Canvas**. Instructions, practice banks, and demo notebooks live on this site — see the course path for the recommended order.

Full grading weights and policies: syllabus.

Overview

Component	Weight	Description
Module 0 (orientation)	3%	Start Here, tech check, introduce-yourself discussion
Quiz A + Quiz B (Modules 1–5)	10%	Low-stakes · open notes · unlimited attempts · **lowest module pair dropped**
Synthesis quizzes (5 modules)	12%	Integration · open notes · **two attempts** · highest score counts
Lab checkpoints	15%	Exercise labs · completion + one rubric criterion per module
Discussions	5%	One community prompt per module · participation, not correctness
Projects 1–5	50%	Applied analyses · write-ups + runnable code
Portfolio package	5%	Module 5 capstone · all five projects + reflection
Total	**100%**	

There are **no midterm or final exams** in this async session. Exam-style reasoning appears in synthesis quizzes and projects instead.

Quizzes

Each module (1–5) has **three Canvas quizzes**:

Quiz	When	Attempts	Practice bank (this site)
Quiz A (Def & Notation)	After Read 1	Unlimited	`quiz-m*-def`
Quiz B (Concept Check)	After Read 2	Unlimited	`quiz-m*-concept`

Quiz	When	Attempts	Practice bank (this site)
Synthesis	After exercise lab (typically Thu)	**Two** (highest counts)	`quiz-m*-synthesis`

All quizzes are **open notes** — use your methods map and R reference.

Practice banks on GitHub include full answer keys. Canvas may draw a random subset each attempt and autograde where possible.

Practice banks by module

Module	Quiz A	Quiz B	Synthesis
1	Def & Notation	Concept Check	Synthesis
2	Def & Notation	Concept Check	Synthesis
3	Def & Notation	Concept Check	Synthesis
4	Def & Notation	Concept Check	Synthesis
5	Def & Notation	Concept Check	Synthesis

Legacy 15-week quiz pages remain on the site for reference but are **not** on the async course path.

Labs

Each module includes a **demo lab** (not graded — walk through on the course site) and an **exercise lab** (submit in Canvas, typically **Wednesday**).

Module	Demo	Exercise (submit)
0	Demo	Exercise
1	Demo, Conditional demo	Exercise
2	Demo	Exercise
3	Demo	Exercise
4	Demo	Exercise
5	Demo	Exercise

Exercise labs are graded on **completion** plus **one visible checkpoint** per module (e.g. a required plot, interpretation sentence, or code cell). Demo labs prepare you for the exercise and for synthesis quiz items.

Submission: Upload your completed `.ipynb` (and any required text) in Canvas. Colab links are on each lab page.

Project milestones

Short **planning checkpoints** (typically **Friday**) help catch direction errors before the full project is due. Submit in Canvas.

Module	Milestone focus
1	Research question; p in context; H_0/H_a; BRP conditions; null-graph draft
2	Population, unit, sampling frame, SRS plan, sample size, CI method
3	Dataset link; four variables typed; plan for two figures + four numeric facts
4	Variables/units; design (RCT vs observational); confounder; method choice; simulation plan + `set.seed()`
5	Team, predictor/response, measurement plan, model draft, valid prediction target

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Projects

Five portfolio projects — each includes a **one-page summary** (PDF or equivalent) and a **runnable R notebook**. Ideal due **Sunday** for Modules 1–4; **Thursday July 2** for Module 5. See syllabus for grace-window policy.

Project	Module	Due (2026)	Link
Project 1 — Testing a Claim	1	Sun **Jun 7**	Project 1
Project 2 — Estimating the Truth	2	Sun **Jun 14**	Project 2
Project 3 — Discovering Patterns	3	Sun **Jun 21**	Project 3
Project 4 — Designing Fair Comparisons	4	Sun **Jun 28**	Project 4
Project 5 — Looking Ahead	5	Thu **Jul 2**	Project 5

Module 1 milestone notebook: Project 1 milestone · `.ipynb`

Project 2 templates: Mean CI notebook · Proportion CI notebook

Project notebook rubric (20 points)

Used across all five projects:

Criterion	Pts	Meets expectations
Runs and is organized	3	Labeled sections; runs top-to-bottom
Correct method and code	4	Method matches question and variable types
Output quality	4	Readable titles, labels, units
Interpretation in context	5	Meaning in problem language, not software jargon
Reproducibility	2	Data source documented; `set.seed()` when simulating
Reflection / limitations	2	At least one limitation or improvement
Total	**20**	

One-page PDF rubrics are on each project page.

Portfolio package (Module 5)

Due **Thursday July 2, 2026** with Project 5 (5% of course grade):

- Index page linking all five projects
- Five one-page summaries + notebook links (polished for submission)
- Brief revision memo if you updated work after feedback
- Short reflection: what you can now do in applied statistics
- Both reference sheets (methods, R reference) submitted or linked

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Discussions

One **community discussion** per module (Modules 0–5) in Canvas. Graded on **participation** (one original post + one thoughtful reply), not statistical correctness. Prompts build connection across the async term — from introducing yourself through place-based and prediction themes.

Grading scale

A: 93–100% · A-: 90–92% · B+: 87–89% · B: 83–86% · B-: 80–82% · C+: 77–79% · C: 73–76% · C-: 70–72% · D: 60–69% · F: 0–59%

A minimum grade of **C-** is required for the course to count toward GE Area B4.

At a glance (module dates)

Module	Dates	Project / capstone due
0 — Start here	Before Jun 1	—
1 — Testing a Claim	Jun 1–7	Sun Jun 7
2 — Estimating the Truth	Jun 8–14	Sun Jun 14
3 — Discovering Patterns	Jun 15–21	Sun Jun 21
4 — Designing Fair Comparisons	Jun 22–28	Sun Jun 28
5 — Looking Ahead	Jun 29–Jul 2	Thu Jul 2 (Project 5 + portfolio)

Details and learning outcomes: course path.
